

Separable Equations

ANSWER KEY



Order and verifying solutions

- 1 State the order of the differential equation $y'' + 3y' - 2y = \sin x$.
Second order (the highest derivative appearing is y'').
- 2 Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.
 $y' = 2e^{2x}$, so $y' - 2y = 2e^{2x} - 2e^{2x} = 0$. Yes, it is a solution.

Recognizing and solving separable equations

- 3 Solve $\frac{dy}{dx} = xy$.
Separate: $\frac{dy}{y} = x dx$. Integrate: $\ln|y| = \frac{x^2}{2} + C$. So $y = Ae^{x^2/2}$.
- 4 Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.
Separate: $\frac{dy}{y^2} = \frac{dx}{x}$. Integrate: $-\frac{1}{y} = \ln x + C$. So $y = \frac{-1}{\ln x + C}$.

Initial value problems

- 5 Solve $\frac{dy}{dx} = 3x^2y$, $y(0) = 2$.
 $\frac{dy}{y} = 3x^2 dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ae^{x^3}$. Since $y(0) = 2$, $A = 2$: $y = 2e^{x^3}$.
- 6 Solve $\frac{dy}{dx} = -2xy$, $y(0) = 5$.
 $\frac{dy}{y} = -2x dx \Rightarrow \ln|y| = -x^2 + C \Rightarrow y = Ae^{-x^2}$. Since $y(0) = 5$, $A = 5$: $y = 5e^{-x^2}$.

Application: radioactive decay and population growth

- 7 A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k .
 $A = 100e^{-kt}$. Then $50 = 100e^{-5k} \Rightarrow e^{-5k} = \frac{1}{2} \Rightarrow -5k = \ln \frac{1}{2} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$ per unit time.
- 8 A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.
 $P(t) = 200e^{0.03t}$, so $P(10) = 200e^{0.3} \approx 269.97$.

Further practice: separable equations

- 9 Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.
Separate: $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$. Integrate: $\arctan y = \arctan x + C$.

10 Solve $y' = y \cos x$, $y(0) = 1$.

$$\frac{dy}{y} = \cos x \, dx \Rightarrow \ln|y| = \sin x + C \Rightarrow y = Ae^{\sin x}. \text{ Since } y(0) = 1, A = 1: y = e^{\sin x}.$$